Docker Installation and OWASP Juice Shop Setup

---

1. Introduction

In modern cybersecurity testing, containerization plays an important role in quickly setting up vulnerable environments. **Docker** allows applications to run in isolated containers, making it easy to deploy and test applications.

In this setup, I used Docker to install and run **OWASP Juice Shop**, which is a deliberately insecure web application designed for practicing web security vulnerabilities.

---

2. Objective

The objective of this setup is:

- To install Docker on the system
- To deploy OWASP Juice Shop using Docker
- To create a safe environment for practicing OWASP Top 10 vulnerabilities

---

3. Docker Installation

Step 1: Update System

```
sudo apt update
```

Step 2: Install Docker

```
sudo apt install docker.io -y
```

Step 3: Start Docker Service

```
sudo systemctl start docker
```

Step 4: Enable Docker at Startup

```
sudo systemctl enable docker
```

Step 5: Verify Installation

docker –version

```
Apps   Places                                   Mar 5  11:37
                                   root@kali: /home/divakar

┌──(root㉿kali)-[/home/divakar]
└─# docker version
Client:
 Version:           27.5.1+dfsg4
 API version:       1.47
 Go version:        go1.24.9
 Git commit:        cab968b3
 Built:             Thu Nov  6 10:43:49 2025
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          27.5.1+dfsg4
  API version:      1.47 (minimum version 1.24)
  Go version:       go1.24.9
  Git commit:       61416484
  Built:            Thu Nov  6 10:43:49 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.7.24~ds1
  GitCommit:        1.7.24~ds1-10
 runc:
  Version:          1.3.3+ds1
  GitCommit:        1.3.3+ds1-2
 docker-init:
  Version:          0.19.0
  GitCommit:
```

This command confirms that Docker is successfully installed.

Step 6: Fix Permission Issue (Optional but Recommended)

sudo usermod -aG docker $USER
newgrp docker

This allows running Docker without using sudo.

---

4. Pulling OWASP Juice Shop Image

After installing Docker, the next step is to download the Juice Shop image.

docker pull bkimminich/juice-shop

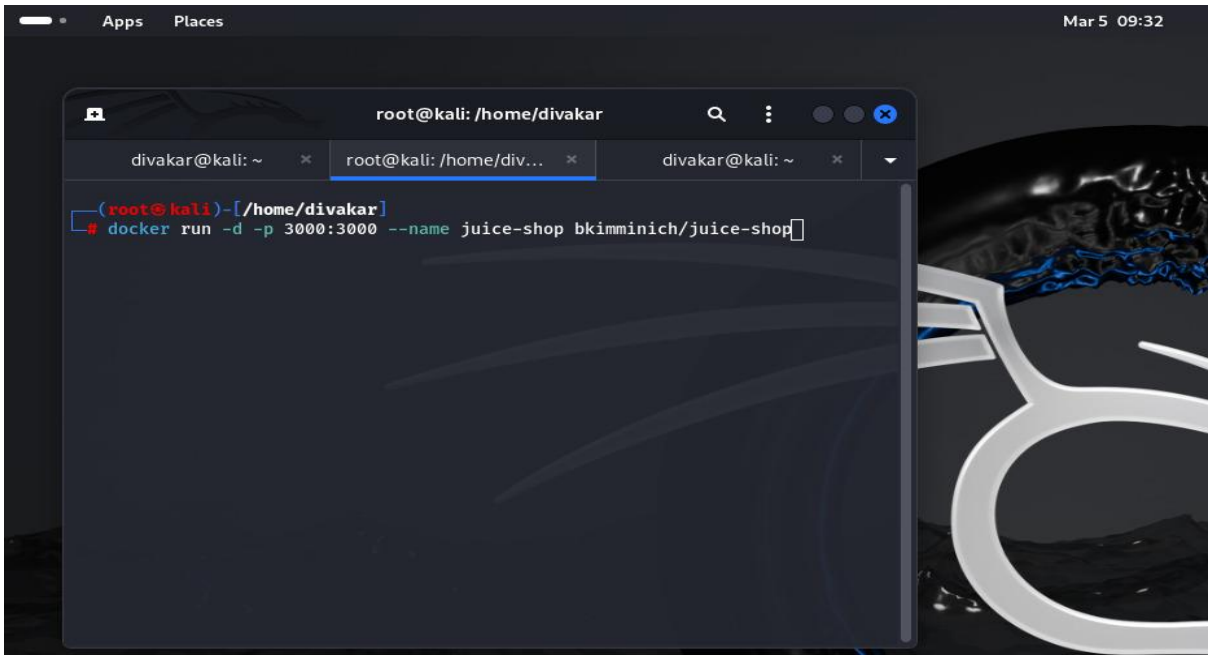
This command downloads the official Juice Shop container from Docker Hub.

---

5. Running OWASP Juice Shop

To start the application:

docker run -d -p 3000:3000 --name juice-shop bkimminich/juice-shop



Explanation:

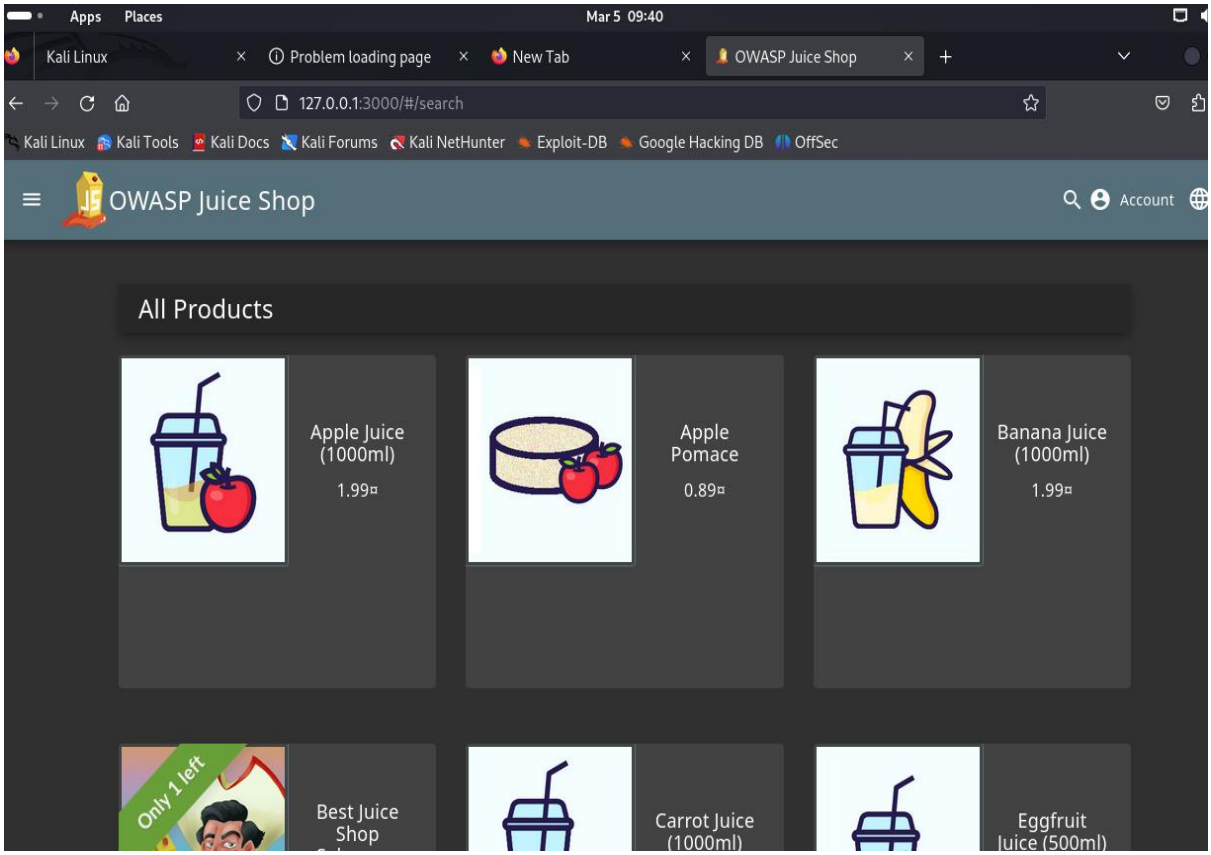
- -d → Runs container in background
- -p 3000:3000 → Maps container port to local system
- --name juice-shop → Assigns container name

---

6. Accessing the Application

Once the container is running, open a browser and navigate to:

http://localhost:3000



This will open the OWASP Juice Shop web application.

---

7. Verifying Running Containers

To check whether Juice Shop is running:

docker ps -a

This command lists all active containers.

---

8. Managing the Container

Stop the container

docker stop juice-shop

Start the container

docker start juice-shop

Remove the container

docker rm -f juice-shop

---

9. Troubleshooting

Issue: Port already in use

docker run -d -p 4000:3000 --name juice-shop2 bkimminich/juice-shop

Access:

http://localhost:4000

Issue: Container not starting

docker logs juice-shop

Issue: Permission denied

sudo usermod -aG docker $USER

---

10. Practical Usage in Security Testing

After successfully running Juice Shop, it can be used for:

- Practicing OWASP Top 10 vulnerabilities
- Testing SQL Injection, XSS, and Authentication flaws
- Learning how to use tools like Burp Suite

---

11. Conclusion

Docker provides a simple and efficient way to deploy vulnerable applications for testing. Using Docker to run **OWASP Juice Shop** helps in creating a controlled environment to learn and practice web security concepts.

This setup is highly useful for beginners and professionals preparing for roles in VAPT and cybersecurity.

---

12. Personal Note

Setting up Juice Shop using Docker helped me understand how real-world applications can be deployed and tested quickly. It also provided hands-on experience in identifying and analyzing vulnerabilities in a safe environment.